

Santander Business Banking Prototype: A Front-End-First Architecture for Demonstrating Paperless Workflows with Embedded Regulatory Compliance

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Abstract. We present a front-end-only prototype of a Santander Business Banking application that demonstrates paperless equivalents of branch- and post-based processes — mandate changes, bulk payments, KYC/KYB, account closures, Making Tax Digital submission, lending, FX, standing orders and Direct Debits, account unlinking, ring-fencing and complaint handling — with the relevant regulatory control embedded in each flow rather than bolted on. The system is a single React component (~6,500 lines) compiled to one self-contained HTML file, runs in any browser with no backend or real data, and ships an immutable audit trail, dual-authorisation with personal-PIN signing, working-day-aware cooling-off, and Confirmation of Payee. We describe the architecture and its deliberate single-file constraint, the entity and mandate models that fork every compliance path, a feature-to-regulation mapping, and the evaluation the prototype supports. We state limitations candidly: the data is fictional, there is no production back-end, and the single-file design is a prototype-stage choice, not a production pattern.

Keywords: paperless banking; dual authorisation; SCA; Confirmation of Payee; FCA Consumer Duty; Making Tax Digital; cooling-off; Direct Debit Guarantee; FCA DISP complaints; single-file React; WCAG 2.1 AA; audit trail.

1 Introduction

Business-banking change requests — adding a signatory, paying wages, closing an account, submitting VAT, cancelling a Direct Debit — have historically meant paper forms, branch visits and posting documents to a processing centre. Each is slow, error-prone and hard to audit. This prototype demonstrates that every one of these can be a guided, self-serve digital workflow in which the regulatory control is part of the flow, not a separate compliance step [1,2].

The prototype is deliberately front-end only. It holds no real customer data, makes no API calls (other than Google Fonts), and is built to be opened on any device for internal presentations and customer research. This scopes the contribution precisely: it is a demonstration of workflow design and embedded compliance, not a production banking system.

1.1 Contributions

- A single-file React architecture (~6,500 lines) that compiles to one self-contained ~890 KB HTML file (~194 KB gzip), readable top-to-bottom and deployable as a static asset.
- Thirteen step-based workflow wizards spanning the key business-banking operations, each forking by entity type and signing rule.
- Embedded controls: dual authorisation with personal-PIN signing (SCA), working-day-aware 24-hour cooling-off, Confirmation of Payee, KYC/KYB, and an FCA DISP-compliant complaint flow.
- A seven-entity model and three-tier mandate engine (Any-1 / Any-2 / All) that drives whether a workflow needs co-signers or a cooling-off period.
- An immutable, actor-and-timestamp audit trail aligned to FCA SYSC 9 — the demonstrability evidence the Senior Managers Regime expects [3].
- A consistent design system (WCAG 2.1 AA contrast, focus-visible, tabular figures) enforced by a six-station pre-commit review process.

2 Background and Regulatory Context

2.1 Strong Customer Authentication and dual authorisation

Payment and mandate actions are authorised under PSD2 Regulatory Technical Standards Art. 97 [4]. The prototype models this as a per-account signing rule and a personal-PIN signing step, with a six-tier step-up matrix escalating to biometric and Voice ID for higher-risk actions.

2.2 Confirmation of Payee and payment safety

Before any payment leaves, a Confirmation of Payee check verifies the payee name against the account, with an explicit override path that cannot be skipped silently (PSR / SI 2019/1215) [5]. APP-fraud reimbursement obligations under PSR PS23/3 frame the complaint and fraud paths [6].

2.3 Cooling-off, Consumer Duty and complaints

High-impact actions carry a 24-working-hour cooling-off period (FCA BCOBS 4A) during which the customer can cancel without penalty; the timer pauses over weekends and bank holidays. The complaint workflow follows FCA DISP — eligible-complainant checks, escalation triage and FOS signposting — consistent with the Consumer Duty obligation to deliver good outcomes [1,7].

2.4 The single-file architecture decision

A central design question separates a defensible prototype from a fragile one. The obvious answer to "make it production-shaped" — split into modules behind a backend — is the wrong one for a demonstration whose value is being read and understood end-to-end. We therefore keep the whole product in one file, accept the trade-off explicitly (§7), and document the production path (§7) rather than pretend the prototype is the product.

3 System Overview and Design Principles

The prototype is organised around three principles. Front-end by default: it runs with no network access and no backend, so it can be demonstrated anywhere. One readable artefact: the entire product is a single React component, so a reviewer can follow every workflow without navigating modules. Compliance in the flow: each control is rendered as part of the workflow that needs it, with the governing regulation named in the UI.

Figure 1. The prototype is one React component compiled to a single HTML file; a shared state tree feeds two render shells (mobile bottom-nav and desktop sidebar) that present the same screens and workflow overlays.

4 Application Architecture

4.1 Single-component structure

App.jsx is one function component. All hooks are declared at the top (120+ state variables) before any logic, because the sub-components — workflow renderers, screens and sheets — are closures and cannot hold their own hooks. A reset function (closeWorkflow) clears all workflow state to prevent ghost state between flows.

4.2 Navigation model

Three state layers compose every view: a tab (home, approve, audit, MTD, statements), a workflow overlay (one of thirteen, or none), and a step index within the active workflow. A viewMode flag switches the mobile and desktop shells while preserving all in-progress state.

4.3 Entity system

A static map configures seven entity types; the active type re-derives labels, required documents, registration numbers, the accounts list and mandate defaults (Table 1).

Entity type	Mandate default	Principal	Companies House	Board minutes
Sole trader	Any-1	Sole trader	No	No
Partnership	Any-1 / Any-2	Partner	No	On rename (RM)

Entity type	Mandate default	Principal	Companies House	Board minutes
Limited company	Any-2	Director	Yes	No
LLP	Any-2	Member	Yes (LLP no.)	No
Registered charity	All	Trustee	Charity Comm.	Yes — all changes
Club	All	Committee member	No	Yes — all changes
Society	All	Committee member	No	Yes — all changes

Table 1. The seven entity types and the configuration each forks. The mandate default drives whether a workflow requires co-signers and a cooling-off period.

4.4 Mandate rules

Each account carries a signing rule — Any-1, Any-2 or All. A helper, `getMandateFor(accounts)`, selects the strictest rule across the selected accounts (All > Any-2 > Any-1). The result determines whether a workflow requires a co-signer and/or triggers the 24-hour cooling-off period.

4.5 Design system

Tailwind utilities with a brand palette (#c8102e red, #faf6ef warm background, stone neutrals) and an inline CSS layer for brand cards and animations. The system enforces WCAG 2.1 AA contrast, a visible focus ring on every interactive element, one primary call-to-action per view, and tabular figures on all monetary amounts [8].

5 Workflows

Thirteen step-based wizards render on top of the main screens; each reads and writes the shared state tree and forks by entity and signing rule. Table 2 catalogues them.

Workflow	Embedded control	Regulatory basis
Account closure	Destination CoP, credit check, cooling-off	FCA BCOBS 4A
Partner-unreachable escalation	4 branches, evidence, restriction	FCA PS22/9
Mandate changes	KYC/KYB, board minutes, signing rule	MLR 2017 / JMLSG
Bulk payments / wages	CoP per payee, dual authorisation	PSR; PSD2 RTS
Business details update	Companies House sync	Companies House
Dormant reactivation	FSCS notice, 12-month threshold	FSCS
MTD VAT submission	Auto-categorise, VAT100, HMRC API	HMRC MTD
Personal/business unlink	Render-tree removal, CRM update	GDPR Art.5(1)(c)
Credit ring-fence	Persisted formal instruction	GDPR Art.5(1)(c)
Pre-approved lending	14-day cooling-off, repayment calc	CCA 1974
International FX payment	Fee disclosure, MLR screening	MLR 2017
Standing orders & Direct Debits	DD Guarantee, CoP on set-up	PSR; SI 2019/1215
Complaint handling	Eligibility, triage, FOS signpost	FCA DISP

Table 2. The thirteen workflow wizards, the control embedded in each, and its regulatory basis.

5.1 Dual authorisation and signing

Pending actions reach a signature queue. Signing uses a personal four-digit PIN by default (biometric and Voice ID remain selectable), framed as SCA. A signed payment then enters a ten-second cancel window before release.

5.2 Cooling-off

The cooling-off timer adds 24 working hours, advancing hour-by-hour and only decrementing on weekday, non-bank-holiday hours; a request started on a Friday therefore executes the following Monday. A live card shows a progress bar, with Cancel and Do-it-now controls.

6 Compliance Mapping

Each control is tied to a specific obligation rather than a vague claim (Table 3).

Control	Regulation	Implementation
Cooling-off	FCA BCOBS 4A	Working-hours aware, cancellable, live progress
Dual authorisation	PSD2 RTS Art.97	Any-1/2/All per account, PIN signing, co-sign window
KYC / KYB	MLR 2017 / JMLSG	GOV.UK One Login Lists 1–3, sanctions, PEP, visa
Audit trail	FCA SYSC 9	7-year, append-only, actor + timestamp per event
Confirmation of Payee	PSR / SI 2019/1215	Pre-payment name check, explicit override
Complaints	FCA DISP	D+3/D+56 deadlines, eligibility, FOS signpost
Data separation	GDPR Art.5(1)(c)	Render-tree removal, CRM unlink, credit ring-fence
Accessibility	WCAG 2.1 AA	Contrast $\geq 4.5:1$, focus-visible, tabular figures

Table 3. Feature-to-regulation mapping. No control is included without a named obligation behind it.

7 Limitations and Path to Production

We state the boundaries plainly, in the spirit of a defensible rather than promotional account.

- Fictional data only. All names, account numbers, balances and amounts are invented; there is no real customer data and no money movement.
- No back-end. The prototype is front-end only; the workflows imply but do not call production services (payments, KYC, HMRC, audit log). A Node.js service layer is the natural Phase-2 step [9].
- Single-file by design. The ~6,500-line single component is intentional for a readable prototype, not a production structure; hardening would migrate to TypeScript and a modular architecture [10].
- Demonstration of controls, not certification. Embedded controls model the correct regulatory behaviour for research and presentation; they are not a substitute for a controlled, tested and independently assured production implementation.

8 Conclusion

The prototype shows that the paper-and-branch processes of business banking can be expressed as guided digital workflows with the governing regulatory control built into each flow, in a single readable, portable artefact. It is a faithful demonstration of design and embedded compliance — and an explicit, documented foundation for a production system, not a stand-in for one.

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